

config.yaml

thresholds and seeds, validated at startup

data/loader.py

UCI 350 fetch + cache

data/processor.py

49 predictors, audit columns

bayes/segment_shrinkage.py

Beta-Binomial empirical Bayes
posterior + 95% CI per segment

models/benchmark.py

3 models, 5-fold stratified CV
ROC-AUC / PR-AUC / KS / Brier / ECE

governance/ - five gates, one verdict per model

fairness

drift

calibration

uncertainty

segment_stability

parity + equalized odds

PSI, train vs test

ECE, MCE, Brier

entropy, bootstrap CI

PD vs credible interval

explainability/shap.py

exact Tree / Linear SHAP

governance/reporter.py

scorecard + model cards

utils/tracking.py

MLflow or JSON run log